No
Date

Jurnal Praktikum
Teknik pengukuran HF
Pengukuran karakteristik RF Mixer

Kelompok : 3 — 12 September 2022

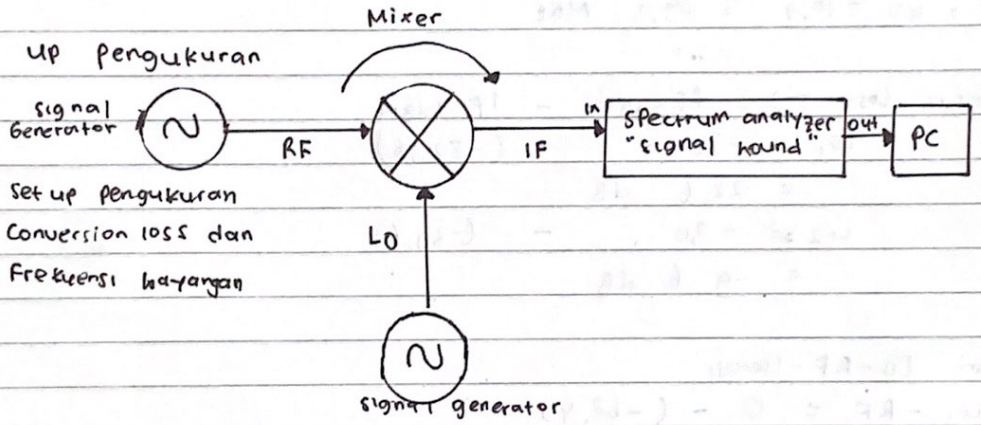
Anggota : Fakhri Nugraha Pratama (2013440l0)
Farhan Mulya Wijaya (2013440ll)
Fauzar Dwipriyandi (2013440l2)

- Tujuan praktikum :

Diakhir percobaan mahasiswa dapat :

1. Memahami karakteristik RF Mixer, sifat dan penggunaannya
2. Mengukur spectrum frekuensi Mixer
3. Mengukur conversion loss
4. Mengukur isolasi RF - LO, IF - LO

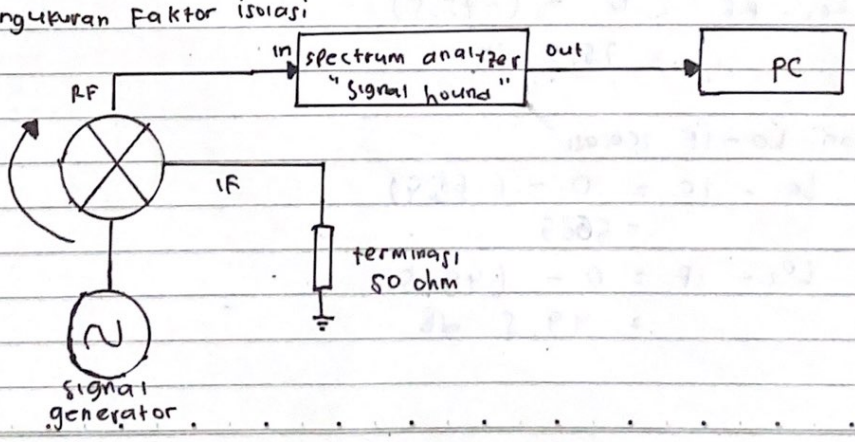
- Set up pengukuran



a. Set up pengukuran Conversion loss dan Frekuensi bayangan

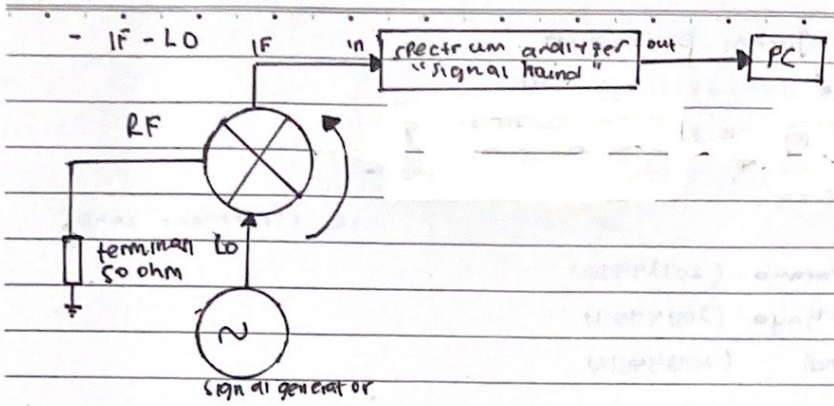
b. Set up pengukuran Faktor isolasi

- RF - LO



KENKO® 36 Lines, 6 mm

No
Date



- Hasil pengukuran

Diketahui : Rf = 100 MHz ⟶ -30 dBm

IF = 10,7 MHz ⟶ 0 dBM

① mencari Lo ⟶ RF ± IF

Lo_1 = 100 + 10,7 = 110,7 MHz

Lo_2 = 100 - 10,7 = 89,3 MHz

② Conversion Loss ⟶ RF (dBm) - IF (dBm)

Lo_1 = -30 - (-52,6)

= 22,6 dB

Lo_2 = -30 - (-39,6)

= 9,6 dB

③ mencari LO - RF Isolasi

Lo_1 - RF = 0 - (-68,4)

= 68,4 dB

Lo_2 - RF = 0 - (-75,5)

= 75,5 dB

④ mencari LO - IF Isolasi

Lo_1 - IF = 0 - (-55,5)

= 55,5

Lo_2 - IF = 0 - (-49,5)

= 49,5 dB

No
Date

⑤ Mencari frekuensi bayangan

$f_{IM} = FRF \pm 2\,FIF$

$FIM_1 = 100 + 21,4$

$= 121,4$ MHz

$FIM_2 = 100 - 21,4$

$= 78,6$ MHz

⑥ nilai data frekuensi bayangan

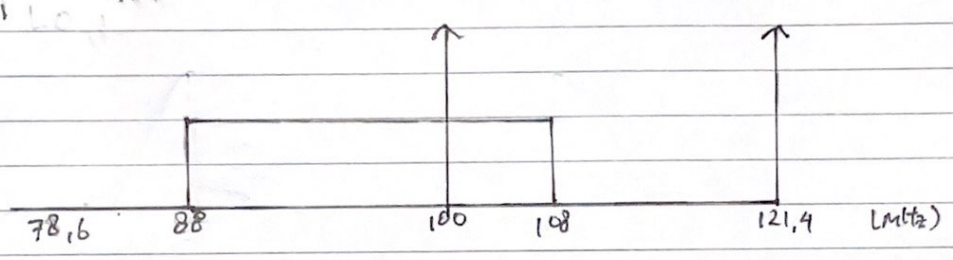
$LO_1 \Rightarrow FIM_1 = -45,3$ dBM → terdapat frekuensi bayangan ✓

$FIM_2 = 82,5$ dBm → tidak muncul frekuensi bayangan ×

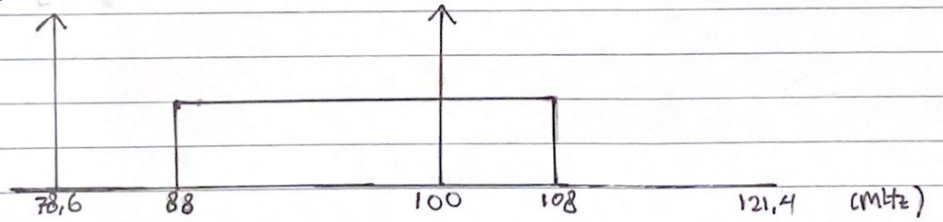
$LO_2 \Rightarrow FIM_1 = 79,0$ dBM → tidak muncul frekuensi bayangan ×

$FIM_2 = -39,7$ dBM → terdapat frekuensi bayangan ✓

LO 1



LO 2



JOYKO® 36 Lines, 6 mm